

Aaron Evans

MECHANICAL ENGINEERING STUDENT

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CATIA V6 & SOLIDWORKS	Prototype Build & Testing	MATLAB & FEA	Drawings & Fixtures
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EXPERIENCE

Tesla | Actuator Mechanical Design Engineer Jun 2026 -- Present
Optimus · Intern · Palo Alto, CA

- Mechanical design engineer on the Optimus actuator team, spanning production component issues, supplier coordination, and design for manufacturing.
- Focused on making actuator components more reliable, manufacturable, and lower-cost.

Urban Solar Corp. | Mechanical Design Co-op Sept -- Dec 2024
Victoria, BC

- Designed PEM installation adapters for prototyping around an off-the-shelf press, avoiding the cost of available commercial setups.
- Supported an IP65 lighting control cabinet release for **2000+** units.
- Created manufacturing drawings, labor time standards, and first-article QC checks.

Professional Components | Mechanical Design Co-op Jan -- Apr 2024
Sidney, BC

- Designed production parts and welding jigs for waterjet, welding, and 3D printing.
- Managed multi-level SOLIDWORKS assemblies and produced part, assembly, and weld drawings.

ENGINEERING PROJECTS

QDD Planetary Gearbox 2026 -- present

- Designed and assembled a **5:1** planetary gearbox for a BLDC motor.
- Measured backlash below a **0.5 deg** requirement.
- Used calibration coupons to set printed bearing and shaft fits before reprinting full parts.

Trailer Hitch Structural Analysis & Redesign 2025 -- 2026

- Built a MATLAB torsion model; crossbar torsion accounted for **94%** of tip deflection.
- Matched V1 hand calcs to FEA within **8.9%**, then cut deflection by **31%**.
- Redesigned the crossbar and gusset package to keep average safety factor at **2.7** under 4G.

Design of Mechanical Elements Shaft Fatigue Solver 2026

- Built an iterative shaft-fatigue sizing tool that resolves interdependent design factors to convergence.

EDUCATION

BEng, Mechanical
University of Victoria
Expected Spring 2028

Coursework: Control Theory, Vibrations, Materials Science, Fluid Mechanics, Machine Elements, Mechanisms

SKILLS

CAD & Modeling
CATIA V6 | SOLIDWORKS

Analysis
MATLAB | FEA | Engineering Drawings | DFM

Manufacturing
3D Printing | Welding | Assembly

Testing
Component Testing | Dyno Setup | Test Fixtures

HANDS-ON

Automotive
Diff rebuild | Trans swap | Suspension mods | Axle swaps | Timing belts | Valve job

Fabrication
MTB bash guard (machined + 3D printed) | Waterjet skid plates | Transformable camping platform